

Hoverfly

Deterministic testing for applications with external APIs

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External dependencies make tests unpredictable

Flaky by nature

Network outages, rate limits, latency and changing payloads.

Hard to reproduce

A failure may depend on a third-party service or a transient condition.

Limited edge cases

Success paths are easy; timeouts, errors and unusual responses are harder.

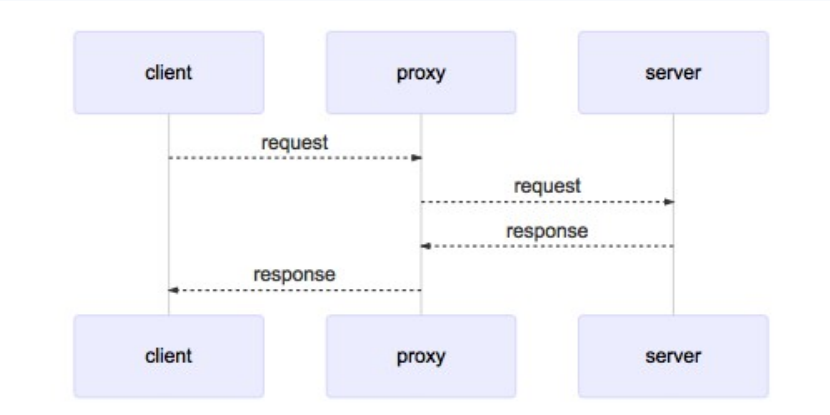
Testing your code should not require the external service to be healthy every time.

Goal: control the dependency while keeping the HTTP interaction realistic.

Hoverfly is a service-virtualization proxy

A lightweight, open-source tool for API simulation and HTTP traffic interception.

Capture real request/response traffic
Replay saved simulations in tests
Match requests to stored responses



Test / app
HTTP client

Hoverfly
Proxy + rules

API
Real or simulated

The test client makes ordinary HTTP calls.

Hoverfly is a controllable stand-in for an API dependency.

Choose the mode for the testing question

CAPTURE

Record real HTTP interactions into a reusable simulation.

SIMULATE

Replay recorded interactions without calling the real API.

MODIFY

Pass traffic through and transform requests and/or responses via middleware.

SPY

Simulate external API only if found a matching request

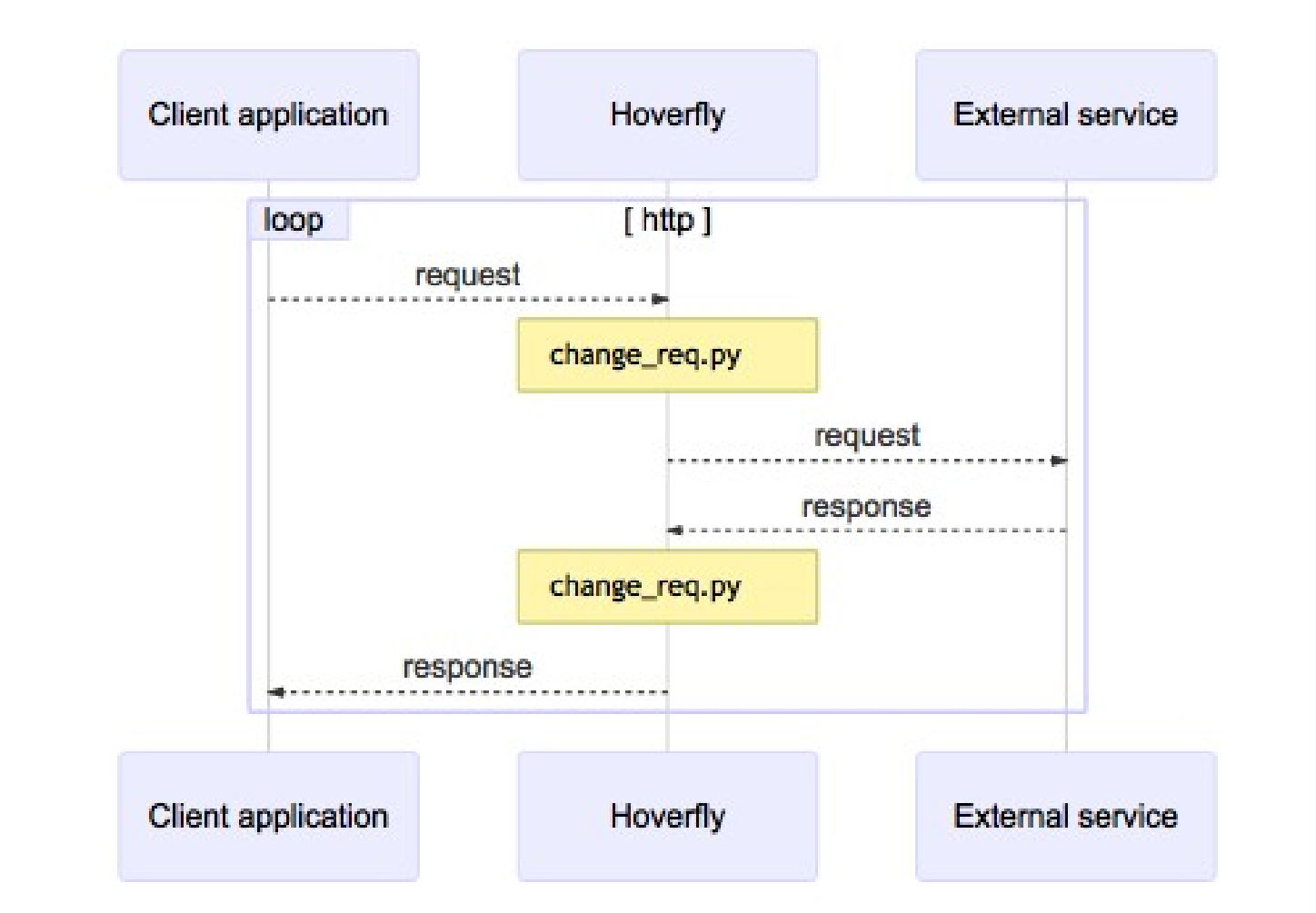
SYNTHESIZE

Create responses dynamically

DIFF

Compare the real external API response vs the stored simulated and store the diff

Modify Mode


Client application
Hoverfly
External service
loop
[http]
request
change_req.py
request
response
change_req.py
response
Client application
Hoverfly
External service

How Hoverfly selects a response

Request

Field	Value
method	GET
destination	www.destination.com

Request Response Pair 3

Field	Matcher Type	Value	Score	Total Score	Matched?
method	exact	GET	+1	2	true
destination	exact	www.destination.com	+1		

Request Response Pair 4

Field	Matcher Type	Value	Score	Total Score	Matched?
method	exact	GET	+1	1	false
destination	exact	www.miss.com	+0		

WINNER: PAIR 3

Both configured matchers succeed, giving it the higher score. Hoverfly returns Pair 3's associated response.

Where Hoverfly fits

Good fit

Integration or end-to-end tests with HTTP dependencies; repeatable regression cases; controlled failures and response variants.

Not a replacement for

All end-to-end tests use cases, contract test, or unit test where a local mock is sufficient.